# From Notches to ABC: Deriving the Optimal Income Tax from Bunching

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**Abstract**

I show that the optimal nonlinear income tax formula be derived from the response to changing the magnitude of a notch in a tax schedule. In particular, I show that the classic ABC optimal nonlinear tax condition *is* derivable from the first-order welfare effect of introducing a notch. This links the literature on bunching at notches to that on optimal income taxation in a way that has some novel pedagogical value relative to existing derivations. In particular, the role of composition effects is made quite stark by using notches as the reform of interest. In addition, I present a second-order condition which demonstrates that the optimal smooth schedule cannot be (locally) improved upon by introducing a notch.

*Keywords:* optimal income taxation, notches, bunching, tax reforms, sufficient statistics.
*JEL:* H21, H23, H24.

## 1 Introduction

The optimal income tax formula has been derived in many ways. Mirrlees (1971) first derived it using mechanism design. Diamond (1998) and Saez (2001) instead perturb the tax schedule and account for who pays more, who earns less, and who is made worse off. Their argument has become the standard sufficient-statistics derivation. Subsequent work develops a general variational calculus, recasts nonlinear income taxation as a linear commodity tax problem, and obtains much of the same content through ordinary optimization arguments (Golosov, Tsyvinski, and Werquin, 2014; Scheuer and Werning, 2016; Gerritsen, 2024). Jacquet and Lehmann (2021) allow taxpayers earning the same income to differ in more than skill. The familiar statistics continue to apply, but must be averaged across the mix of taxpayers at each income.

In this paper I provide another derivation based on bunching at a notch. Unlike the smooth tax reforms typically used in optimal tax theory derivations, notches are common in real tax systems (Blinder and Rosen, 1985; Slemrod, 2013). They also produce a familiar, and intuitive empirical pattern reflecting the behavioral response to taxation: a mass of taxpayers at the threshold and a nearly empty interval immediately above it (Kleven and Waseem, 2013; Kleven, 2016).

Consider adding a small notch to a smooth income tax schedule. Taxpayers immediately above the threshold give up their last few dollars of earnings and move down to it. Why does the width of this bunching band scale with the square root of the notch? The reason is that the marginal buncher starts from an interior optimum. The first derivative of her objective is zero there, so giving up $s$ dollars of income creates a loss that is quadratic in $s$. Avoiding a notch of size $\kappa$, by contrast, creates a gain that is linear in $\kappa$. Indifference therefore requires $s^2$ to be proportional to $\kappa$.

This square-root rule provides a particularly simple route to the behavioral term in the optimal tax formula. The number of bunchers is proportional to the width of the band and is therefore of order $\sqrt{\kappa}$. The average buncher also gives up an amount of income of order $\sqrt{\kappa}$. Multiplying the two produces a first-order loss of taxable income. More formally, the total income displaced within a taste group is the area of a triangle: one half times the local density times the squared band width. Substituting the marginal-buncher indifference condition turns this area into the compensated taxable-income response multiplied by the density at the threshold. The marginal tax rate on the income given up then produces the standard behavioral-revenue term.

The argument makes composition effects visible. Fix a taste type. There is one marginal skill, and therefore one sharp endpoint to that taste group's missing-income interval. The post-notch density does not recover gradually within a homogeneous taste cell. It returns at one cutoff. The gradual recovery observed in the aggregate is instead created by staggered cutoffs across cells with different compensated responses. This distinction matters because different objects sample the cells with different weights. Bunching mass is proportional to the square root of the elasticity, while taxable income displaced is proportional to the elasticity itself. The latter is exactly the composition-weighted mean that appears in the formula of Jacquet and Lehmann (2021).

This main result is not wholy without precedent. Dudek, Dudek, and Walczyk (2023) introduce the same thought experiment with quadratic utility and one dimension of heterogeneity, and recover the optimal tax condition of Diamond (1998). I extend the derivation to a fully modern ABC formula in the spirit of Jacquet and Lehmann (2021).

At an optimal smooth schedule, the ordinary $O(\kappa)$ effect of the notch vanishes. The next terms are cubic in the bunching-band width. Because that width is $O(\sqrt{|\kappa|})$, the planner's objective changes at order $|\kappa|^{3/2}$ and has no finite second derivative at zero. I derive the leading coefficient. With no within-income taste heterogeneity, it is a curvature condition involving preferences, the tax schedule, and the drift of the compensated response across taxpayers. With multidimensional tastes, one additional covariance appears because the notch samples taste cells in proportion to $\sqrt{\epsilon}$ rather than local density alone. The result gives a condition under which no sufficiently small notch of either sign can improve on an optimal smooth schedule.

There is also a useful connection to the empirical notch literature. The homogeneous square-root inversion is the shrinking-notch counterpart of the structural marginal-buncher equation in Kleven and Waseem (2013) and the corrected reduced-form approximation in Kleven (2018). With heterogeneous responses, total bunching identifies a square-root moment, while the first moment of the actual missing-density profile equals total taxable income displaced and therefore recovers the mean compensated response needed for optimal-tax accounting. Hong (2024), in particular, develops a finite-notch density-ratio method to estimate an elasticity distribution using a control group. The point here is narrower: the same recovery profile shows how heterogeneous discrete movements assemble the behavioral term in the ABC formula.

Section 2 introduces the taxpayers, the planner, and the notch. It derives the square-root rule, converts the bunching response into total taxable income displaced, and obtains the ABC formula. Section 3 studies the experiment beyond first order and gives the notch-robustness condition. Section 4 briefly records how the same indifference calculation changes when a taxpayer's two candidate incomes remain separated. Section 5 concludes. The appendices provide the formal two-sided expansion, the derivation of the $|\kappa|^{3/2}$ term, and the separated-optimum calculation.

## 2 One income, many tastes

### 2.1 Taxpayer choice and local curvature

Taxpayers differ in skill $w$ and a taste type $\theta$ of arbitrary dimension. Conditional on tastes, skill has distribution $F(\cdot \mid \theta)$ and density $f(\cdot \mid \theta)$, while tastes follow a measure $\mu$. A taxpayer of type $(w, \theta)$ has smooth preferences $u(c, z; w, \theta)$ over consumption $c$ and taxable income $z$. Utility rises with consumption and falls with income. The baseline tax schedule $\mathcal{T}$ is smooth and satisfies $1 - \mathcal{T}'(z) > 0$ at every relevant income. A taxpayer earning $z$ consumes $z - \mathcal{T}(z)$, plus lump-sum income $\rho$. I retain $\rho$ because the notch acts like a change in lump-sum income for taxpayers who remain above it.

Given the tax schedule, define the taxpayer's objective and policy function by

$$\begin{aligned} G(z; w, \theta, \rho) &\equiv u(z - \mathcal{T}(z) + \rho, z; w, \theta), \\ Z(w, \theta, \rho) &\in \arg\max_z G(z; w, \theta, \rho). \end{aligned} \tag{1}$$

At an interior optimum, the first- and second-order conditions are

$$0 = G'(z) = (1 - \mathcal{T}'(z))u_c + u_z, \tag{2}$$

$$0 > G''(z) = -\mathcal{T}''u_c + (1 - \mathcal{T}')^2 u_{cc} + 2(1 - \mathcal{T}')u_{cz} + u_{zz}. \tag{3}$$

The first line defines the policy function. The second says that a taxpayer earns until the value of the last dollar she keeps, $(1 - \mathcal{T}')u_c$, equals the effort cost $-u_z$. The third line gives the curvature of her objective at this choice. Notice, this curvature determines how strongly the taxpayer is attached to her chosen income. When the objective is sharply curved, she gives up income reluctantly. When it is nearly flat, she can be moved farther at little cost.

I impose single crossing in the form

$$(1 - \mathcal{T}')u_{cw} + u_{zw} > 0.$$

Thus, a move up the schedule is worth more to a taxpayer with greater skill. Differentiating the first-order condition with respect to skill, lump-sum income, and the net-of-tax rate with utility held fixed gives

$$\begin{aligned} \frac{\partial Z}{\partial w} &= -\frac{(1 - \mathcal{T}')u_{cw} + u_{zw}}{G''}, \qquad\qquad \eta \equiv \frac{\partial Z}{\partial \rho} = -\frac{(1 - \mathcal{T}')u_{cc} + u_{cz}}{G''}, \\ \epsilon &\equiv \frac{1 - \mathcal{T}'}{z} \frac{\partial Z^c}{\partial (1 - \mathcal{T}')} = -\frac{(1 - \mathcal{T}')u_c}{zG''}. \end{aligned} \tag{4}$$

Single crossing makes $\partial Z / \partial w$ positive, mapping the skill distribution into the income distribution. The income effect $\eta$ is nonpositive when consumption is a normal good. The final object is the compensated elasticity of taxable income along the nonlinear schedule. Apart from the observable scaling by income and the net-of-tax rate, it is marginal utility of consumption divided by the curvature of the taxpayer's objective. A taxpayer responds strongly when a dollar is especially valuable to her or when her objective is relatively flat.[1]

[1]Because $G''$ contains $\mathcal{T}''$, this is a response along the nonlinear schedule rather than against a linearized budget. Jacquet and Lehmann (2021) call these total responses and provide the conversion to their linearized counterparts.

## 2.2 The planner, the notch, and the bunchers

The planner values a taxpayer's utility through a smooth increasing transformation $\Phi(\cdot; w, \theta)$ and values public funds at $\lambda$, the marginal value of public funds. I write the planner's Lagrangian and the resulting welfare weight as

$$\mathcal{L} \equiv \iint \left[ \mathcal{T}(Z(w, \theta, 0)) + \frac{\Phi(u(w, \theta); w, \theta)}{\lambda} \right] f(w \mid \theta) \, \mathrm{d}w \, \mu(\mathrm{d}\theta), \qquad g(w, \theta) \equiv \frac{\Phi_u u_c}{\lambda}. \tag{5}$$

The Lagrangian is measured in dollars of public funds. Thus, $g$ is the marginal social value of a dollar in the taxpayer's hands in the same units as tax revenue.

Fix a location $z^\star$ on the smooth schedule and add a surcharge $\kappa$ above it:

$$T(z; \kappa) = \mathcal{T}(z) + \kappa \mathbf{1}\{z > z^\star\}. \tag{6}$$

A single number describes this reform. Marginal tax rates remain unchanged, while every taxpayer above $z^\star$ owes an additional $\kappa$.

Fix a taste type $\theta$ and let $w^\star(\theta)$ denote the skill whose baseline optimal income is $z^\star$. Taxpayers below the threshold are unaffected. Taxpayers well above it pay the surcharge and adjust their interior incomes through the income effect in (4). Taxpayers immediately above the threshold respond differently. Their last few dollars of income were nearly worthless at the margin, yet these dollars now trigger the full surcharge. They instead give up the income and locate at $z^\star$.

I call the taxpayer at the upper edge of this band the marginal buncher. Let her skill be $w^I(\kappa, \theta)$ and let $z^I(\kappa, \theta)$ be her interior optimum after paying the surcharge. She is exactly indifferent between paying the surcharge at this interior choice and retreating to the threshold:

$$u(z^I - \mathcal{T}(z^I) - \kappa, z^I; w^I, \theta) = u(z^\star - \mathcal{T}(z^\star), z^\star; w^I, \theta). \tag{7}$$

The remaining derivation follows from this condition. Its left side is utility at the interior optimum after paying the surcharge. Its right side is utility at the threshold after giving up income and avoiding it. Define

$$\Delta_\theta(\kappa) \equiv z^I(\kappa, \theta) - z^\star.$$

Each taste cell has its own marginal buncher and therefore its own band width $\Delta_\theta$.

Figure 1 shows why multidimensional heterogeneity need not produce diffuse bunching within a taste cell. Conditional on $\theta$, there is one marginal skill and one sharp endpoint. The aggregate density recovers gradually because different cells return at different points. The taste cells in the figure are ordered by their local elasticities only for display; $\theta$ itself may be arbitrarily multidimensional.[2]

[2] The figure suppresses the bounded $O(\kappa)$ shifts of upper-branch taxpayers. The bunching bands are $O(\sqrt{\kappa})$, so these shifts do not affect the leading local geometry.

### 2.3 The square-root rule

The indifference condition has a simple local expansion. For the marginal buncher, define the objective on the surcharged upper branch by

$$G_\kappa(z) \equiv u(z - \mathcal{T}(z) - \kappa, z; w^I, \theta).$$

Her interior choice satisfies $G'_\kappa(z^I) = 0$, while (7) says $G_\kappa(z^I) = G_0(z^\star)$. Adding and subtracting $G_\kappa(z^\star)$ gives

$$G_\kappa(z^I) - G_\kappa(z^\star) = G_0(z^\star) - G_\kappa(z^\star). \tag{8}$$

The right side is the utility value of avoiding the surcharge,

$$G_0(z^\star) - G_\kappa(z^\star) = u_c^\star \kappa + o(\kappa).$$

The left side is the gain from being at the upper-branch optimum rather than retreating to the threshold. Since the first derivative is zero at $z^I$, a second-order expansion gives

$$G_\kappa(z^I) - G_\kappa(z^\star) = \frac{1}{2}[-G_\theta^{\star\prime\prime}]\Delta_\theta^2 + o(\Delta_\theta^2).$$

Here and below, a star indicates evaluation at the type-$\theta$ taxpayer earning $z^\star$ on the baseline schedule. Appendix A verifies that the marginal buncher's skill and both candidate bundles converge to this point, so the displayed expansion may be evaluated there.

Equating the two sides and using (4) yields

$$\Delta_\theta(\kappa)^2 = \frac{2\kappa z^\star \epsilon_\theta^\star}{1 - \mathcal{T}'(z^\star)} + o(\kappa). \tag{9}$$

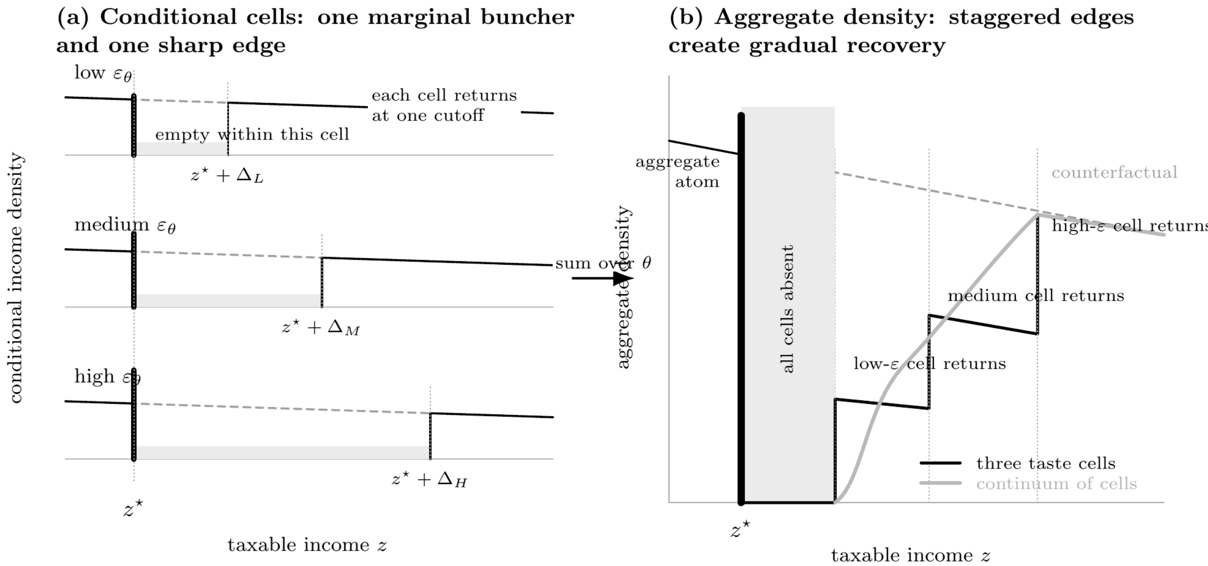
This is the square-root rule. Avoiding the notch creates a first-order gain, while retreating from an optimum creates a second-order loss. The bunching-band width must therefore be of order $\sqrt{\kappa}$.

It is useful to express the same calculation in money-metric units. If a taxpayer gives up $s$ dollars of income from an interior optimum, her local loss is

$$\ell_\theta(s) = \frac{1 - \mathcal{T}'(z^\star)}{2z^\star \epsilon_\theta^\star} s^2 + o(s^2). \tag{10}$$

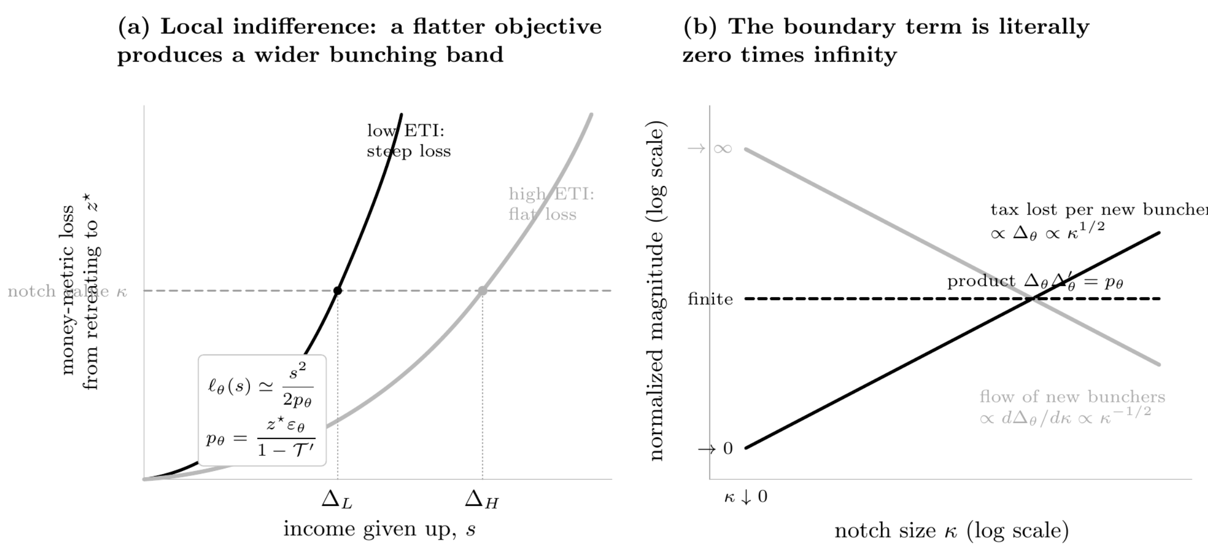
The marginal buncher satisfies $\ell_\theta(\Delta_\theta) = \kappa$ to first order. A flatter objective means a larger compensated elasticity, a flatter loss curve, and a wider band. Figure 2 depicts this comparison and the equivalent zero-times-infinity interpretation. In the figure, $p_\theta$ is only a shorthand for $z^\star \epsilon_\theta / [1 - \mathcal{T}'(z^\star)]$.

Notice, the coefficient in (9) is the compensated taxable-income response. A smooth increase in the local net-of-tax rate tilts the taxpayer's objective, while a notch gives a fixed reward once she crosses




$$h_0(z^\star + s) - h_\kappa(z^\star + s) \;\approx\; \int h_\theta(z^\star)\,\mathbf{1}\{\Delta_\theta(\kappa) > s\}\,d\mu(\theta)$$

**Figure 1:** Conditional hard cutoffs and aggregate recovery. Within each taste cell, the post-notch density is empty up to one cutoff and then returns. Summing cells with different cutoffs produces a staircase in the discrete example and a smooth recovery in the continuum.




$$\ell_\theta(\Delta_\theta) = \kappa \implies \frac{d\Delta_\theta}{d\kappa} = \frac{p_\theta}{\Delta_\theta}, \qquad -\mathcal{T}' h_\theta \Delta_\theta \frac{d\Delta_\theta}{d\kappa} \longrightarrow -\frac{\mathcal{T}'}{1-\mathcal{T}'} z^\star h_\theta \varepsilon_\theta$$

**Figure 2:** The square-root rule and the boundary interpretation. Panel (a) equates the notch with the quadratic loss from retreating. Panel (b) differentiates the resulting displaced-income area: the income lost per new buncher vanishes, the flow of new bunchers diverges, and their product remains finite.

the threshold. The two reforms have different geometry, but both reveal the same local curvature.

For the revenue calculation, it is convenient to measure the bunching band in the baseline income distribution. Let

$$\overline{\Delta}_\theta(\kappa) \equiv Z\big(w^I(\kappa,\theta),\theta,0\big) - z^\star. \tag{11}$$

The perturbed upper optimum includes the surcharge's lump-sum income effect, while the baseline choice does not. Therefore

$$\overline{\Delta}_\theta = \Delta_\theta + \eta_\theta^\star\kappa + o(\kappa), \qquad \overline{\Delta}_\theta^2 = \Delta_\theta^2 + o(\kappa). \tag{12}$$

The income-effect adjustment is $O(\kappa)$, while the band itself is $O(\sqrt{\kappa})$. It does not change the leading squared width.

### 2.4 The displaced-income triangle

Let $h(z \mid \theta)$ denote the baseline income density within taste cell $\theta$. A taxpayer whose baseline income is $z^\star + s$, for $0 < s < \overline{\Delta}_\theta$, moves to the threshold and gives up $s$ dollars of taxable income. Total income displaced in the cell is therefore

$$\begin{aligned} A_\theta(\kappa) &\equiv \int_0^{\overline{\Delta}_\theta(\kappa)} s\, h(z^\star + s \mid \theta)\,\mathrm{d}s \\ &= \frac{1}{2} h(z^\star \mid \theta)\overline{\Delta}_\theta(\kappa)^2 + o(\kappa) \\ &= \kappa\,\frac{z^\star \epsilon_\theta^\star}{1-\mathcal{T}'(z^\star)} h(z^\star \mid \theta) + o(\kappa). \end{aligned} \tag{13}$$

The first line defines an area. Under a locally flat density, it is a triangle with height equal to the bunching mass and base equal to the band width. The second line is the local area formula. The third substitutes the square-root rule.

The government loses the marginal tax rate on this displaced income. Thus, the first-order buncher contribution in cell $\theta$ is

$$-\mathcal{T}'(z^\star)A_\theta(\kappa) = -\kappa\frac{\mathcal{T}'(z^\star)}{1-\mathcal{T}'(z^\star)} z^\star\epsilon_\theta^\star h(z^\star \mid \theta) + o(\kappa). \tag{14}$$

This is the behavioral-revenue term before aggregating over tastes.

Why do the bunchers' own welfare losses not appear in (14)? A buncher at distance $s$ loses $O(s^2)$ in money metric. Integrating this loss over a band of width $\overline{\Delta}_\theta$ gives $O(\overline{\Delta}_\theta^3) = O(\kappa^{3/2})$. Tax curvature and density drift across the band have the same order. The direct notch payment omitted for the bunchers is $\kappa$ times a mass of order $\sqrt{\kappa}$, which is again $O(\kappa^{3/2})$. These terms vanish after dividing by $\kappa$, although they become the leading terms once the first-order condition is imposed in Section 3.

The hierarchy can be summarized as

$$\begin{aligned}
\underbrace{\overline{\Delta}_\theta}_{\text{band width}} &= O(\kappa^{1/2}), & \underbrace{h_\theta \overline{\Delta}_\theta}_{\text{bunching mass}} &= O(\kappa^{1/2}), \\
\underbrace{\tfrac{1}{2} h_\theta \overline{\Delta}_\theta^2}_{\text{income displaced}} &= O(\kappa), & \underbrace{h_\theta \overline{\Delta}_\theta^3}_{\text{cubic terms}} &= O(\kappa^{3/2}).
\end{aligned} \tag{15}$$

This is also the zero-times-infinity calculation in Figure 2. Differentiating the triangle gives

$$\frac{\mathrm{d}A_\theta}{\mathrm{d}\kappa} \simeq h(z^\star \mid \theta)\Delta_\theta \frac{\mathrm{d}\Delta_\theta}{\mathrm{d}\kappa}.$$

Equation (9) implies that $\Delta_\theta \to 0$ and $\mathrm{d}\Delta_\theta / \mathrm{d}\kappa \to \infty$, but

$$\Delta_\theta \frac{\mathrm{d}\Delta_\theta}{\mathrm{d}\kappa} \longrightarrow \frac{z^\star \epsilon_\theta^\star}{1 - \mathcal{T}'(z^\star)}. \tag{16}$$

The area proof and the moving-boundary proof are the same mathematics. The former evaluates the triangle and divides by $\kappa$; the latter differentiates it.

Figure 3 gives a distribution-space version. The gap between the post-notch and counterfactual CDFs is the cumulative amount of missing density remaining above each distance $s$. Its area is total taxable income displaced. Panel (b) records the equivalent fact that the squared band width is locally linear in the notch, with slope determined by the compensated elasticity.

## 2.5 Aggregation and the ABC formula

To aggregate the cell-level calculation, let

$$\widehat{h}(z) \equiv \int h(z \mid \theta)\, \mu(\mathrm{d}\theta), \qquad \widehat{H}(z) \equiv \int_0^z \widehat{h}(s)\,\mathrm{d}s \tag{17}$$

be the population income density and distribution. For any taxpayer statistic $x$, define its composition average at income $z$ by

$$\widehat{x}(z) \equiv \frac{\int x(z,\theta) h(z \mid \theta)\, \mu(\mathrm{d}\theta)}{\widehat{h}(z)}. \tag{18}$$

These are the hats of Jacquet and Lehmann (2021). The dimension of taste heterogeneity enters only through the mix of taxpayers represented at each income.

Taxpayers who remain above the bunching band contribute the familiar mechanical, welfare, and income-effect terms. The surcharge raises one dollar from each of them. Its private cost is valued at the welfare weight $g$, while the lump-sum income response changes smooth tax revenue by $-\mathcal{T}'\eta$. Their contribution is therefore

$$\kappa \int_{z^\star}^{\infty} \left[1 - \widehat{g}(z) - \mathcal{T}'(z)\widehat{\eta}(z)\right] \widehat{h}(z)\,\mathrm{d}z + o(\kappa). \tag{19}$$

Strictly, the lower edge of this group is the top of the bunching band rather than $z^\star$. Moving it down to the threshold adds a strip with mass $O(\sqrt{\kappa})$ whose contribution per taxpayer is $O(\kappa)$. The resulting error is $O(\kappa^{3/2})$.

Combining (14) and (19), then integrating across tastes, gives

$$\begin{aligned}\mathcal{L}(\kappa) - \mathcal{L}(0) = \kappa \Bigg\{ &\int_{z^\star}^{\infty} \left[1 - \widehat{g}(z) - \mathcal{T}'(z)\widehat{\eta}(z)\right] \widehat{h}(z)\,\mathrm{d}z \\ &- \frac{\mathcal{T}'(z^\star)}{1-\mathcal{T}'(z^\star)} z^\star \widehat{\epsilon}(z^\star)\widehat{h}(z^\star) \Bigg\} + o(\kappa). \end{aligned} \tag{20}$$

The second line follows because

$$\int \epsilon_\theta^\star h(z^\star \mid \theta)\,\mu(\mathrm{d}\theta) = \widehat{\epsilon}(z^\star)\widehat{h}(z^\star).$$

High-elasticity taste cells supply both more bunchers and larger movements. The two square-root effects combine to produce the ordinary density-weighted mean elasticity.

Equation (20) was derived for a positive surcharge. A subsidy notch produces the mirrored response. Taxpayers immediately below the threshold climb to it, and taxpayers already above it receive the subsidy as a lump-sum transfer. Appendix A shows that the same coefficient multiplies negative $\kappa$. Thus, the derivative at zero is two-sided:

$$\begin{aligned}\mathcal{L}_\kappa(0; z^\star) = &\int_{z^\star}^{\infty} \left[1 - \widehat{g}(z) - \mathcal{T}'(z)\widehat{\eta}(z)\right] \widehat{h}(z)\,\mathrm{d}z \\ &- \frac{\mathcal{T}'(z^\star)}{1-\mathcal{T}'(z^\star)} z^\star \widehat{\epsilon}(z^\star)\widehat{h}(z^\star). \end{aligned} \tag{21}$$

At an optimum, a notch of either sign cannot improve the planner's objective at any threshold. Setting (21) equal to zero and rearranging gives

$$\frac{\mathcal{T}'(z^\star)}{1-\mathcal{T}'(z^\star)} = \underbrace{\frac{1}{\widehat{\epsilon}(z^\star)}}_{A} \cdot \underbrace{\frac{1-\widehat{H}(z^\star)}{z^\star \widehat{h}(z^\star)}}_{B} \cdot \underbrace{\frac{\int_{z^\star}^{\infty} \left[1 - \widehat{g}(z) - \mathcal{T}'(z)\widehat{\eta}(z)\right] \widehat{h}(z)\,\mathrm{d}z}{1-\widehat{H}(z^\star)}}_{C}. \tag{22}$$

Equation (22) is the ABC formula of Diamond (1998) and Saez (2001). The $A$ factor is the inverse average compensated elasticity at the threshold. Less responsive taxpayers create less displaced income, allowing the threshold to support a higher marginal rate. The $B$ factor compares the upper tail, whose taxpayers all pay the mechanical dollar, with the income-weighted density at the threshold, whose taxpayers create the distortion. A thin density beneath a large upper tail supports a higher rate. The $C$ factor is the average social value of taking a dollar from taxpayers above the threshold: one dollar collected, less their welfare weight, and less the revenue returned through income effects.

This route begins with the empirical bunching picture. A taxpayer gives up income to avoid a notch; a taste cell contributes a sharp missing interval; and the aggregate response is the sum of the resulting displaced-income triangles. The formal calculation uses the same sufficient statistics as a smooth perturbation, but reaches them through behavior that can be seen in the income distribution. Notice as well that the elasticity is not introduced as a primitive. Equation (4) constructs it from the marginal utility and curvature that also determine the marginal buncher's indifference condition.

### 2.6 Relation to empirical notch moments

The same geometry clarifies what different features of a bunching distribution summarize. Let $d_\kappa(s)$ denote the local missing density at distance $s > 0$ above the threshold, after separating the buncher-induced hole from the bounded upper-branch income-effect shift. Define its zeroth and first moments by

$$M_{0,\kappa} \equiv \int_0^\infty d_\kappa(s)\,\mathrm{d}s, \qquad M_{1,\kappa} \equiv \int_0^\infty s\,d_\kappa(s)\,\mathrm{d}s. \tag{23}$$

The first object is the ordinary missing mass. The second has units of taxpayer-dollars and is total taxable income displaced. Equivalently, if

$$D_\kappa(s) \equiv \int_s^\infty d_\kappa(r)\,\mathrm{d}r$$

is the excess-CDF curve shown in Figure 3, Fubini's theorem gives

$$M_{1,\kappa} = \int_0^\infty D_\kappa(s)\,\mathrm{d}s. \tag{24}$$

Let expectations with a star use the composition weights among taxpayers earning $z^\star$. Equations (9) and (13) imply

$$M_{0,\kappa} = \widehat{h}(z^\star)\sqrt{\frac{2\kappa z^\star}{1-\mathcal{T}'(z^\star)}}\,\widehat{\sqrt{\epsilon}}(z^\star) + o(\sqrt{\kappa}), \tag{25}$$

$$M_{1,\kappa} = \kappa\frac{z^\star}{1-\mathcal{T}'(z^\star)}\widehat{h}(z^\star)\widehat{\epsilon}(z^\star) + o(\kappa). \tag{26}$$

Thus, total bunching weights taste cells by $\sqrt{\epsilon_\theta}$, while total income displaced weights them by $\epsilon_\theta$. In particular,

$$\frac{1-\mathcal{T}'(z^\star)}{\kappa z^\star \widehat{h}(z^\star)} M_{1,\kappa} \longrightarrow \widehat{\epsilon}(z^\star). \tag{27}$$

In the homogeneous case, this statement contains no additional identifying content. The excess-CDF curve is a triangle, and total bunching already reveals its width. Substituting that width into the corrected reduced-form expression of Kleven (2018) gives the same factor of one half as (13). With heterogeneous responses, however, applying the homogeneous formula to the average width yields

$$\frac{1-\mathcal{T}'(z^\star)}{2\kappa z^\star}\left(\frac{M_{0,\kappa}}{\widehat{h}(z^\star)}\right)^2 \longrightarrow \left[\widehat{\sqrt{\epsilon}}(z^\star)\right]^2. \tag{28}$$

The difference from the ABC-relevant mean is

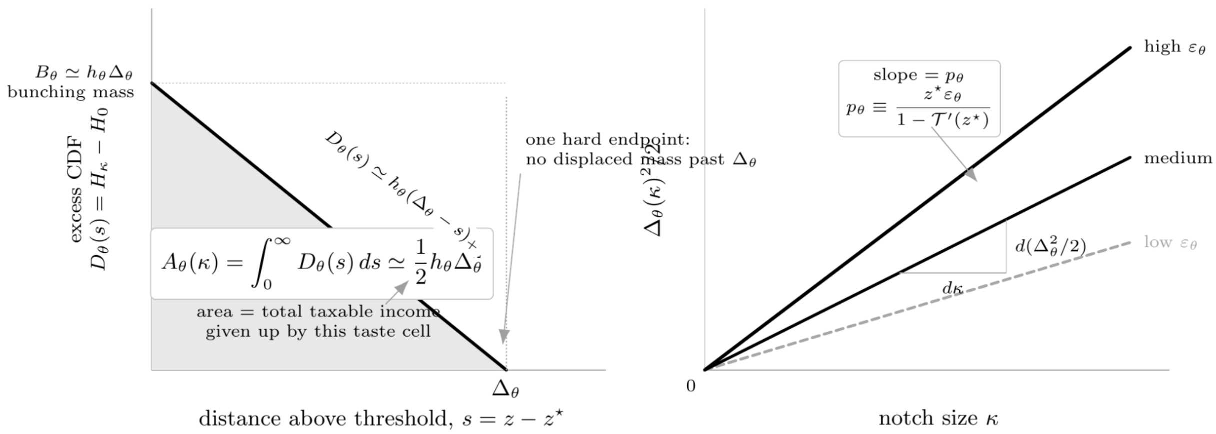
$$\widehat{\epsilon}(z^\star) - \left[\widehat{\sqrt{\epsilon}}(z^\star)\right]^2 = \widehat{\mathrm{Var}}_{z^\star}(\sqrt{\epsilon}). \tag{29}$$

The actual recovery curve supplies this variance correction. Replacing it with one synthetic triangle removes precisely the information created by heterogeneity.

This observation is closely related to existing identification arguments. Kleven and Waseem (2013) use a finite-notch marginal-buncher condition to map a response interval into an elasticity, while Kleven (2018) supplies the corrected reduced-form approximation. Hong (2024) goes further in a setting with a control density, using pointwise treated-to-control density ratios to estimate a conditional elasticity distribution. I do not develop a competing estimation procedure here. The role of (27) is to connect the recovery profile to the particular average response that enters optimal-tax accounting. In an empirical application, optimization frictions, diffuse destinations, finite-notch corrections, and the smooth shift of taxpayers who remain above the notch would all have to be treated explicitly.[3]

[3] The upper-branch shift is $O(\kappa)$ and is negligible for the rescaled local recovery shape, which lives on an $O(\sqrt{\kappa})$ window. It can nevertheless contribute at order $\kappa$ if the CDF difference is integrated over the entire upper tail. The empirical object corresponding to (26) is therefore the local buncher-induced missing mass, or an adjusted object from which the smooth upper-branch shift has been removed.

**(a) One taste cell: the excess-CDF triangle turns bunching mass into taxable-income loss** **(b) The ETI is the local slope of squared bunching-band width**




$$\underbrace{\frac{d}{d\kappa}\frac{\Delta_\theta(\kappa)^2}{2}}_{\text{slope in panel (b)}} \longrightarrow \frac{z^\star \varepsilon_\theta}{1 - \mathcal{T}'(z^\star)}, \qquad \frac{dA_\theta}{d\kappa} \longrightarrow h_\theta(z^\star)\frac{z^\star \varepsilon_\theta}{1 - \mathcal{T}'(z^\star)}$$

**Figure 3:** The first moment of the missing-density profile. Panel (a) shows that the area under the excess-CDF curve equals total taxable income displaced within a taste cell. Panel (b) shows the equivalent square-root inversion. In the schematic, $p_\theta$ denotes $z^\star \epsilon_\theta / [1 - \mathcal{T}'(z^\star)]$.

# 3 Robustness of a smooth optimum to notches

Suppose the smooth schedule satisfies (22) at every threshold. The first-order effect in (20) is then zero. Does this imply that a small notch lowers welfare? Not necessarily. The first-order condition is silent because all of the terms that distinguish a local maximum from a local minimum were discarded as $o(\kappa)$.

The moment ladder in (15) tells us where to look. The first omitted terms are cubic in the bunching-band width. They arise because the paying group begins above the band rather than at the threshold, because the bunchers themselves lose utility, because the tax schedule and density drift across the band, and because the leading square-root rule requires a finite-width correction. Since $\Delta_\theta^3 = O(|\kappa|^{3/2})$, these terms dominate the ordinary $O(\kappa^2)$ curvature associated with a smooth reform.

## 3.1 The common-taste benchmark

Begin with the case in which the optimality condition can be imposed taste by taste, as with one effective dimension of heterogeneity. Define the robustness coefficient

$$K_\theta(z) \equiv \frac{1}{\epsilon_\theta(z)} - \frac{z}{1-\mathcal{T}'(z)}\left[\mathcal{T}''(z) + \frac{\mathcal{T}'(z)}{\epsilon_\theta(z)}\frac{\partial(1/\epsilon_\theta(z))/\partial w}{\partial Z(w,\theta,0)/\partial w}\right]. \tag{30}$$

Appendix B expands the planner's objective through the third power of the bunching gap. In the common-taste benchmark, the result is

$$\mathcal{L}(\kappa) - \mathcal{L}(0) = -\frac{\sqrt{2}}{3}\sqrt{\frac{z^\star}{1-\mathcal{T}'(z^\star)}}h(z^\star)\epsilon(z^\star)^{3/2}K(z^\star)|\kappa|^{3/2} + o(|\kappa|^{3/2}). \tag{31}$$

The inverse elasticity in (30) measures how sharply the taxpayer is attached to her optimum. The term involving $\mathcal{T}''$ removes curvature supplied by the tax schedule itself, leaving the curvature of preferences along the budget direction. The final term accounts for the fact that the marginal bunchers are drawn from nearby skills whose compensated responses need not be the same. A positive $K$ means that these forces make the cubic coefficient in (31) negative, so a notch lowers the planner's objective.

A convenient sufficient condition is easy to state. The first two terms of $K$ are positive when utility is strictly concave along the budget direction. If marginal tax rates are nonnegative and the inverse elasticity is nonincreasing in skill, the final adjustment is weakly positive as well. These conditions are sufficient, not necessary. Equation (30) is useful because it shows which local primitives determine the sign when they fail.

### 3.2 Composition across tastes

With multidimensional tastes, the aggregate first-order condition need not hold within every taste cell. To record the discrepancy, define the type-$\theta$ residual

$$r_\theta(z) \equiv \frac{1}{h(z \mid \theta)} \left\{ \left[1 - g_\theta(z) - \mathcal{T}'(z)\eta_\theta(z)\right] h(z \mid \theta) + \left[\frac{\mathcal{T}'(z)}{1-\mathcal{T}'(z)} z\epsilon_\theta(z) h(z \mid \theta)\right]' \right\}. \tag{32}$$

Differentiating the aggregate first-order condition across neighboring thresholds gives

$$\int r_\theta(z) h(z \mid \theta)\, \mu(\,\mathrm{d}\theta) = 0. \tag{33}$$

Thus, the common schedule balances these residuals under the ordinary composition weights at each income, even though a particular cell may be left locally over- or under-taxed.

The notch does not sample cells with those weights. Its band is wider by a factor of $\sqrt{\epsilon_\theta}$. Appendix B shows that this reweighting is the only additional term needed after the type-specific algebra is collected:

$$\begin{aligned}\mathcal{L}(\kappa) - \mathcal{L}(0) = &-\frac{\sqrt{2}}{3}\sqrt{\frac{z^\star}{1-\mathcal{T}'(z^\star)}}\,\widehat{h}(z^\star)\left[\widehat{\epsilon^{3/2}K}(z^\star) + 2\widehat{\mathrm{Cov}}(\sqrt{\epsilon}, r)(z^\star)\right]|\kappa|^{3/2} \\ &+ o(|\kappa|^{3/2}).\end{aligned} \tag{34}$$

Here the hats use the composition weights in (18), and

$$\widehat{\mathrm{Cov}}(x, y)(z) \equiv \widehat{xy}(z) - \widehat{x}(z)\widehat{y}(z).$$

Since $\widehat{r}(z) = 0$, the covariance term is simply the $\sqrt{\epsilon}$-weighted average residual. In the positive-covariance case, relatively elastic cells are also the cells with positive residuals. Their wider bands receive more weight, which strengthens the welfare loss from the notch. A negative covariance can instead overturn an otherwise positive average robustness coefficient.

The $|\kappa|^{3/2}$ term implies that the usual second derivative does not exist as a finite number. Dividing (34) by $\kappa^2/2$ gives the extended second directional derivative

$$\lim_{\kappa \to 0} \frac{2[\mathcal{L}(\kappa) - \mathcal{L}(0)]}{\kappa^2} = \begin{cases} -\infty, & \widehat{\epsilon^{3/2}K}(z^\star) + 2\widehat{\mathrm{Cov}}(\sqrt{\epsilon}, r)(z^\star) > 0, \\ +\infty, & \widehat{\epsilon^{3/2}K}(z^\star) + 2\widehat{\mathrm{Cov}}(\sqrt{\epsilon}, r)(z^\star) < 0. \end{cases} \tag{35}$$

Smooth reforms move the objective at order $\kappa^2$. If the bracketed term is positive, the smooth schedule is locally protected against a notch at $z^\star$. If it is negative, a small notch strictly improves the objective even though the first-order condition does not detect it.

**Proposition 1** (No notch at the optimum)**.** *Under the regularity conditions in Appendix B, let the smooth schedule* $\mathcal{T}$ *satisfy* (22) *at every threshold and let* $\widehat{h}(z^\star) > 0$*. Then* (34) *and* (35) *hold. If*

*$K_\theta(z^\star) > 0$ for every taste represented at $z^\star$ and*

$$\widehat{\mathrm{Cov}}(\sqrt{\epsilon}, r)(z^\star) \geq 0,$$

*every sufficiently small notch of either sign at $z^\star$ strictly lowers the planner's Lagrangian.*

The proposition concerns one local discontinuity at one threshold. It does not characterize every discontinuous schedule near $\mathcal{T}$. Finitely many separated notches add when their bunching bands do not overlap. Thresholds whose distance from one another shrinks with $\kappa$ require a joint calculation, as do reforms that attract taxpayers to a remote alternative optimum. The result is therefore a local test of notch robustness, rather than a global sufficiency theorem for the smooth schedule.

## 4 Separated optimal incomes

The square-root calculation applies when the two candidate incomes merge as the notch vanishes. The same indifference argument has a second case. Suppose a taxpayer is indifferent between two separated optimal incomes, $z^- < z^+$. This can arise with multidimensional heterogeneity even when each branch is locally well behaved (Bergstrom and Dodds, 2021). A small reform may then cause the taxpayer to jump from one optimum to the other.

Fix a taste type and let $m(\theta)$ denote the skill at a tied boundary. Place the threshold strictly between $z^-$ and $z^+$, so the surcharge reaches only the upper branch. The boundary taxpayer satisfies

$$u(z^+ - \mathcal{T}(z^+) - \kappa, z^+; m, \theta) = u(z^- - \mathcal{T}(z^-), z^-; m, \theta). \tag{36}$$

Differentiating the optimized branch values and applying the envelope theorem gives

$$\left.\frac{\mathrm{d}m}{\mathrm{d}\kappa}\right|_{\kappa=0} = \frac{u_c^+}{u_w^+ - u_w^-}. \tag{37}$$

The corresponding boundary contribution to revenue is

$$J(\theta) = \left[\mathcal{T}(z^-) - \mathcal{T}(z^+)\right] f(m \mid \theta) \left.\frac{\mathrm{d}m}{\mathrm{d}\kappa}\right|_{\kappa=0}. \tag{38}$$

Welfare boundary terms cancel because the marginal taxpayer is indifferent between the two locations.

Compare (38) with the buncher calculation in Section 2. When the two locations merge, the tax-liability gap vanishes and the taxpayer flow diverges. Their product becomes the intensive behavioral term in (21). When the locations remain separated, both objects are finite, producing an ordinary extensive-margin correction. Summing (38) over isolated tied boundaries and tastes gives the correction of Bergstrom and Dodds (2021). Appendix C records the isolation and domination conditions needed for the discontinuous reform.

This observation is useful mainly as a boundary of the main argument. The notch experiment treats intensive and extensive responses through the same object—an indifference boundary—but the square-root geometry, the displaced-income triangle, and the $|\kappa|^{3/2}$ result belong to the case in which the candidate locations merge.

# 5 Conclusion

The standard optimal income tax formula is usually derived by perturbing a smooth tax schedule. This paper reaches the same formula by adding a notch and following the taxpayers who bunch at it. The key step is a Taylor expansion around the marginal buncher's interior optimum. Avoiding the notch is worth an amount proportional to $\kappa$, while retreating from the optimum creates a loss proportional to the square of the income given up. The bunching-band width is therefore proportional to $\sqrt{\kappa}$.

This square-root rule turns the observed bunching response into the behavioral term in the ABC formula. Within each taste cell, total income displaced is the area of a triangle. Its base and height are each proportional to $\sqrt{\kappa}$, so its area is first order in the notch. Substituting the indifference condition produces the compensated taxable-income response multiplied by the density at the threshold. Aggregating across tastes yields the composition-weighted ETI of Jacquet and Lehmann (2021). The smooth aggregate recovery above a notch can therefore be understood as a mixture of sharp type-specific cutoffs, and the average elasticity in the tax formula as the sum of the discrete movements behind those cutoffs.

The first-order formula is known. The contribution of the notch experiment is partly interpretive: it connects the sufficient statistic to the spike, missing region, and excess-CDF area familiar from empirical bunching work. It also clarifies why total bunching and total income displaced average heterogeneous responses differently. Existing work already develops the finite-notch identification of elasticity distributions, so I do not treat this connection as a new estimation method.

The main new result concerns what remains after the ABC condition has eliminated the first-order term. A small notch changes the planner's objective at order $|\kappa|^{3/2}$. The leading coefficient contains a local robustness coefficient and, with multidimensional tastes, one covariance that captures the notch's $\sqrt{\epsilon}$ reweighting of the taste distribution. When that coefficient has the appropriate sign, an optimal smooth schedule is locally protected not only against smooth perturbations, but also against a small discontinuity at the threshold.

The derivation is deliberately narrow. It studies one taxable-income choice while allowing unrestricted taste heterogeneity behind that choice. This is enough to make the composition mechanism visible and to obtain the robustness result without the notation required for a fully multidimensional choice problem. The separated-optimum calculation shows how the same indifference boundary becomes an extensive-margin term when the candidate locations do not merge. Other discontinuous reforms, overlapping notch bands, and realistic optimization frictions remain separate problems.

# A The first-order expansion and two-sided limit

## A.1 Regularity

I maintain the following conditions in a neighborhood of the threshold. Preferences are three times continuously differentiable, with $u_c > 0 > u_z$, and the baseline objective has a strict interior optimum with $G'' < 0$. The single-crossing expression $(1 - \mathcal{T}')u_{cw} + u_{zw}$ is strictly positive. The schedule is twice continuously differentiable and satisfies $1 - \mathcal{T}' > 0$. Conditional on each taste, the baseline policy $Z(w, \theta, 0)$ is continuously differentiable and strictly increasing in skill, with a conditional income density continuous at $z^\star$. Baseline and threshold optima are separated from any remote alternative by a locally uniform utility gap that survives sufficiently small notches. Finally, the type-specific remainder terms below have integrable envelopes, allowing the expansions to be integrated over tastes.

The global-isolation condition matters because a local indifference equation does not by itself prevent a taxpayer from choosing a distant third location. The condition is automatic in a globally concave one-dimensional problem, but must otherwise be imposed.

## A.2 A positive surcharge

Fix a taste type and suppress $\theta$. Let $w^I(\kappa)$ and $z^I(\kappa)$ denote the marginal buncher and her upper-branch optimum. Define

$$G_\kappa(z; w) \equiv u(z - \mathcal{T}(z) - \kappa, z; w, \theta).$$

The indifference condition gives

$$G_\kappa(z^I; w^I) - G_\kappa(z^\star; w^I) = G_0(z^\star; w^I) - G_\kappa(z^\star; w^I) = \kappa u_c^\star + o(\kappa). \tag{39}$$

The right side vanishes. Strict curvature and global isolation therefore force $z^I \to z^\star$ and $w^I \to w^\star$. Expanding the left side around its maximum $z^I$ gives

$$\frac{1}{2}(-G^{\star\prime\prime})(z^I - z^\star)^2 + o((z^I - z^\star)^2).$$

Using $u_c/(-G'') = z\epsilon/(1 - \mathcal{T}')$ proves

$$(z^I - z^\star)^2 = \frac{2z^\star\epsilon^\star}{1 - \mathcal{T}'(z^\star)}\kappa + o(\kappa). \tag{40}$$

In particular, $z^I - z^\star = O(\sqrt{\kappa})$ and $\kappa/(z^I - z^\star) \to 0$. No differentiability of the boundary at $\kappa = 0$ or application of l'Hôpital's rule is needed.

Let

$$\overline{\Delta} = Z(w^I, \theta, 0) - z^\star.$$

Since $z^I = Z(w^I, \theta, -\kappa)$, differentiability of the interior policy in lump-sum income gives

$$\overline{\Delta} = (z^I - z^\star) + \eta^\star \kappa + o(\kappa), \qquad \overline{\Delta}^2 = (z^I - z^\star)^2 + o(\kappa). \tag{41}$$

Change variables from skill to baseline income within the bunching band. Its revenue contribution is

$$\begin{aligned}\int_0^{\overline{\Delta}} [\mathcal{T}(z^\star) - \mathcal{T}(z^\star + s)] h(z^\star + s)\, \mathrm{d}s &= -\frac{1}{2}\mathcal{T}'(z^\star) h(z^\star) \overline{\Delta}^2 + o(\kappa) \\ &= -\kappa \frac{\mathcal{T}'(z^\star)}{1 - \mathcal{T}'(z^\star)} z^\star \epsilon^\star h(z^\star) + o(\kappa).\end{aligned} \tag{42}$$

A buncher at distance $s$ leaves an optimum, so her money-metric utility loss is $O(s^2)$. Integrating across the band gives $O(\overline{\Delta}^3) = O(\kappa^{3/2})$. The direct notch payment avoided by the bunchers and all density- or tax-curvature corrections have the same or smaller order.

For a taxpayer who remains above the band, the surcharge acts as lump-sum income of $-\kappa$. The envelope theorem and the interior policy response give a contribution

$$\kappa[1 - g - \mathcal{T}'\eta] + o(\kappa).$$

Replacing the lower boundary of this group by $w^\star$ incurs an error $O(\kappa^{3/2})$. Changing variables from skill to income and adding (42) yields the positive-side version of (20). Uniform domination permits integration over tastes.

### A.3 A negative notch

Write $a = -\kappa > 0$ and include the threshold in the subsidized region:

$$T(z; -a) = \mathcal{T}(z) - a\mathbf{1}\{z \geq z^\star\}.$$

Taxpayers immediately below the threshold climb to it to receive the transfer. Let $z^J < z^\star$ be the interior optimum of the marginal claimant. The mirrored indifference condition is

$$u(z^\star - \mathcal{T}(z^\star) + a, z^\star; w^J, \theta) = u(z^J - \mathcal{T}(z^J), z^J; w^J, \theta).$$

The same flat-maximum expansion gives

$$(z^\star - z^J)^2 = \frac{2z^\star\epsilon^\star}{1-\mathcal{T}'(z^\star)}a + o(a). \tag{43}$$

The claimants raise their smooth-tax payments by the marginal tax rate times the income they add. Their total smooth-tax contribution is therefore

$$+a\frac{\mathcal{T}'(z^\star)}{1-\mathcal{T}'(z^\star)}z^\star\epsilon^\star h(z^\star) + o(a),$$

which equals $\kappa$ times the behavioral coefficient in (20). The subsidy paid to the claimants is $a$ times a mass of order $\sqrt{a}$, and their welfare and curvature terms are also $O(a^{3/2})$. Taxpayers already above the threshold contribute $-a[1-g-\mathcal{T}'\eta] + o(a)$. Hence

$$\mathcal{L}(-a) - \mathcal{L}(0) = -a\,\mathcal{L}_\kappa(0; z^\star) + o(a),$$

which establishes the two-sided expansion.

# B The $|\kappa|^{3/2}$ expansion

This appendix derives (34). Fix a taste type through the first three steps and suppress its index. Assume that $u$ and $\mathcal{T}$ are four times continuously differentiable locally and that $h$ is continuously differentiable. Baseline optima remain strict and globally isolated, and the expansion remainders have integrable envelopes across tastes. Objects without arguments are evaluated at the threshold taxpayer and baseline bundle. A prime on $u_c$, $\epsilon$, or $h$ denotes a derivative along the baseline schedule, across taxpayers at their chosen incomes. A derivative with respect to $w$ instead holds the bundle fixed.

Let

$$\Pi \equiv -\left.\frac{\mathrm{d}G''}{\mathrm{d}z}\right|_{\text{along the schedule}} = -G''' - \frac{G''_w}{\partial Z/\partial w}. \tag{44}$$

The first equality defines the drift of objective curvature across the taxpayers supplying the bunchers. The second separates a within-taxpayer third derivative from an across-taxpayer component.

## B.1 The band width to third order

Let $\Delta = z^I - z^\star$ for a positive notch. Expanding the marginal-buncher indifference condition one order beyond (40) gives

$$\kappa = \frac{1-\mathcal{T}'}{2z^\star\epsilon}\Delta^2 + A_3\Delta^3 + o(\Delta^3), \tag{45}$$

where

$$A_3 = \frac{\Pi}{2u_c} + \frac{G'''}{6u_c} - \frac{(1-\mathcal{T}')u_{cw}}{2z^\star \epsilon u_c(\partial Z/\partial w)}. \tag{46}$$

The final term appears because the utility price of the surcharge is evaluated at the marginal buncher's skill rather than at $w^\star$. It vanishes under quasilinear utility. Equation (45) implies

$$\Delta = \sqrt{\frac{2z^\star\epsilon}{1-\mathcal{T}'}\kappa} + O(\kappa), \qquad \Delta^2 = \frac{2z^\star\epsilon}{1-\mathcal{T}'}[\kappa - A_3\Delta^3] + o(\Delta^3). \tag{47}$$

## B.2 The components of the planner's objective

The baseline width of the bunching band is

$$\overline{\Delta} = \Delta + \eta\kappa + o(\kappa).$$

Changing variables from skill to baseline income, the bunchers' smooth-tax revenue changes by

$$\begin{aligned}\int_0^{\overline{\Delta}} [\mathcal{T}(z^\star) - \mathcal{T}(z^\star + s)]h(z^\star + s)\,\mathrm{d}s = &-\frac{\mathcal{T}'h}{2}\Delta^2 - \mathcal{T}'\eta h\,\kappa\Delta \\ &-\left(\frac{\mathcal{T}''h}{6} + \frac{\mathcal{T}'h'}{3}\right)\Delta^3 + O(\Delta^4).\end{aligned} \tag{48}$$

A buncher at distance $s$ loses

$$\ell(s) = \frac{1-\mathcal{T}'}{2z^\star\epsilon}s^2 + O(s^3)$$

in money metric. Valuing this loss at the local welfare weight and integrating gives

$$-\frac{(1-\mathcal{T}')gh}{6z^\star\epsilon}\Delta^3 + O(\Delta^4). \tag{49}$$

Finally, taxpayers above the band contribute $\kappa(1-g-\mathcal{T}'\eta)$ each to first order. Relative to integrating this term from $z^\star$, the displaced lower boundary removes

$$\kappa(1-g-\mathcal{T}'\eta)h\Delta + O(\kappa^2). \tag{50}$$

Using

$$\kappa\Delta = \frac{1-\mathcal{T}'}{2z^\star\epsilon}\Delta^3 + o(\Delta^3)$$

and the inversion in (47), the income-effect terms cancel. The terms linear in $\kappa$ assemble into the first-order derivative, leaving

$$\begin{aligned}\mathcal{L}_\theta(\kappa) - \mathcal{L}_\theta(0) =& \kappa\mathcal{L}_{\kappa,\theta}(0) \\ & - \Bigg[ \underbrace{\frac{(1-\mathcal{T}')(1-g)h}{2z^\star\epsilon}}_{\text{shrunken paying group}} + \underbrace{\frac{(1-\mathcal{T}')gh}{6z^\star\epsilon}}_{\text{bunchers' utility losses}} \\ & + \underbrace{\frac{\mathcal{T}''h}{6} + \frac{\mathcal{T}'h'}{3}}_{\text{drift across the band}} - \underbrace{\frac{z^\star\mathcal{T}'\epsilon}{1-\mathcal{T}'}A_3h}_{\text{width correction}} \Bigg]\Delta^3 + o(\Delta^3). \end{aligned} \tag{51}$$

Each term has a direct interpretation. The paying group begins one band width above the threshold; the bunchers bear the quadratic displacement loss; the tax schedule and density change across the band; and converting the band width into the statutory notch brings in $A_3$.

### B.3 Using the neighboring first-order conditions

At the optimum, the aggregate first-order condition is zero at every threshold. Differentiating it with respect to the threshold gives (33), with $r_\theta$ defined by (32). Substitute

$$(1-g)h = r_\theta h + \mathcal{T}'\eta h - \left[\frac{\mathcal{T}'}{1-\mathcal{T}'}z\epsilon h\right]'$$

into the bracket of (51). The remaining simplification uses three identities. First,

$$\left[\ln\left(\frac{z\epsilon}{1-\mathcal{T}'}\right)\right]' = \frac{u_c'}{u_c} - \frac{\Pi}{-G''}. \tag{52}$$

Second, $u_c' = -\eta G'' + u_{cw}/(\partial Z/\partial w)$. Third, at a fixed bundle,

$$\frac{\partial(1/\epsilon)}{\partial w} = -\frac{u_{cw} + [z\epsilon/(1-\mathcal{T}')]G_w''}{\epsilon u_c}. \tag{53}$$

Together with $\Pi + G''' = -G_w''/(\partial Z/\partial w)$, these identities cancel the income effect, density drift, elasticity drift, and every third derivative in the consumption and income directions. After collecting terms,

$$6\left(\frac{z^\star\epsilon}{1-\mathcal{T}'}\right)^{3/2} \times [\text{bracket in (51)}] = \sqrt{\frac{z^\star}{1-\mathcal{T}'}}\left[\epsilon^{3/2}K + 2\sqrt{\epsilon}\,r_\theta\right]h, \tag{54}$$

where $K$ is given by (30). Since

$$\Delta^3 = \left(\frac{2z^\star\epsilon}{1-\mathcal{T}'}\right)^{3/2}\kappa^{3/2} + o(\kappa^{3/2}),$$

the type-$\theta$ contribution beyond first order is

$$-\frac{\sqrt{2}}{3}\sqrt{\frac{z^\star}{1-\mathcal{T}'}}\left[\epsilon_\theta^{3/2}K_\theta + 2\sqrt{\epsilon_\theta}\,r_\theta\right] h(z^\star \mid \theta)\kappa^{3/2} + o(\kappa^{3/2}). \tag{55}$$

### B.4 Aggregation and the subsidy side

Integrate (55) over tastes. The threshold and marginal tax rate are common across cells. The $K$ terms become $\widehat{\epsilon^{3/2}K}\widehat{h}$, while

$$\int \sqrt{\epsilon_\theta}\,r_\theta h(z^\star \mid \theta)\,\mu(\,\mathrm{d}\theta) = \widehat{h}(z^\star)\widehat{\mathrm{Cov}}(\sqrt{\epsilon}, r)(z^\star)$$

because (33) implies $\widehat{r}(z^\star) = 0$. The aggregate first-order term is zero by (22), yielding (34) for $\kappa > 0$.

For a subsidy, write $a = |\kappa|$ and let $\Delta^- = z^\star - z^J$ be the marginal claimant's gap. The third-order indifference expansion is

$$a = \frac{1-\mathcal{T}'}{2z^\star\epsilon}(\Delta^-)^2 - A_3(\Delta^-)^3 + o((\Delta^-)^3). \tag{56}$$

The drift runs in the opposite direction, reversing the cubic width correction. Claimant revenue is obtained by integrating $\mathcal{T}(z^\star) - a - \mathcal{T}(z^\star - s)$ against $h(z^\star - s)$, while claimant welfare is $g[a - \ell(s)]$ in money metric. The orientations of these terms reverse together. Collecting them produces the same bracket as (51), with $(\Delta^-)^3$ in place of $\Delta^3$. Hence the aggregate expression depends on $|\kappa|^{3/2}$ and (34) holds for either sign. Proposition 1 follows immediately.

**A useful sufficient condition.** The first two terms of $K$ satisfy

$$\frac{1}{\epsilon} - \frac{z^\star\mathcal{T}''}{1-\mathcal{T}'} = -\frac{z^\star}{1-\mathcal{T}'}\frac{(1-\mathcal{T}')^2u_{cc} + 2(1-\mathcal{T}')u_{cz} + u_{zz}}{u_c}. \tag{57}$$

They are positive when utility is strictly concave along the budget direction. Therefore $K > 0$ whenever this condition holds, $\mathcal{T}' \geq 0$, and $\partial(1/\epsilon)/\partial w \leq 0$. Under quasilinear utility $u = c - \psi(z; w)$, the expression reduces to

$$K = \frac{z^\star}{1-\mathcal{T}'}\left[\psi_{zz} + \mathcal{T}'\frac{\psi_{wzz}}{\psi_{wz}}\right]. \tag{58}$$

## C Separated branch optima

Fix a taste type and suppose that at skill $m$ the baseline objective has exactly two relevant strict local maxima, $z^- < z^+$. Assume that the two branch values are separated from every other candidate optimum by a uniform gap; that the threshold lies strictly between the branches; that the reform is therefore constant in a neighborhood of each branch; and that

$$u_w^+ - u_w^- > 0.$$

The final inequality is the transversality supplied by single crossing. Assume also that the resulting boundary terms are dominated by an integrable, summable envelope when there are multiple isolated ties.

Let $V^+(w,\kappa)$ and $V^-(w,\kappa)$ be the optimized branch values. The tied boundary $m(\kappa)$ solves

$$\Psi(w,\kappa) \equiv V^+(w,\kappa) - V^-(w,\kappa) = 0.$$

The branch value functions are differentiable by the envelope theorem even though the reform direction is discontinuous at the threshold, since the threshold lies outside the two neighborhoods being optimized over. The implicit function theorem gives

$$\frac{\mathrm{d}m}{\mathrm{d}\kappa} = -\frac{\Psi_\kappa}{\Psi_w} = \frac{u_c^+}{u_w^+ - u_w^-},$$

which is (37).

To see the contribution to the planner's objective, write its type-specific integral as a sum over smooth policy branches with endpoints $m_i(\kappa)$. Leibniz's rule produces a boundary term equal to the difference between the two branches' tax payments, multiplied by the density and boundary flow. The corresponding welfare boundary terms cancel because the boundary taxpayer has the same utility on both branches. For a straddling tie, the result is

$$[\mathcal{T}(z^-) - \mathcal{T}(z^+)]f(m \mid \theta)\frac{\mathrm{d}m}{\mathrm{d}\kappa}.$$

Summing over isolated ties and integrating over tastes gives the correction reported in Section 4. If both branches lie above the threshold, the reform reaches both and the numerator of the flow becomes $u_c^+ - u_c^-$. A boundary entirely below the threshold does not move.

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